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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Wave on String

English Medium

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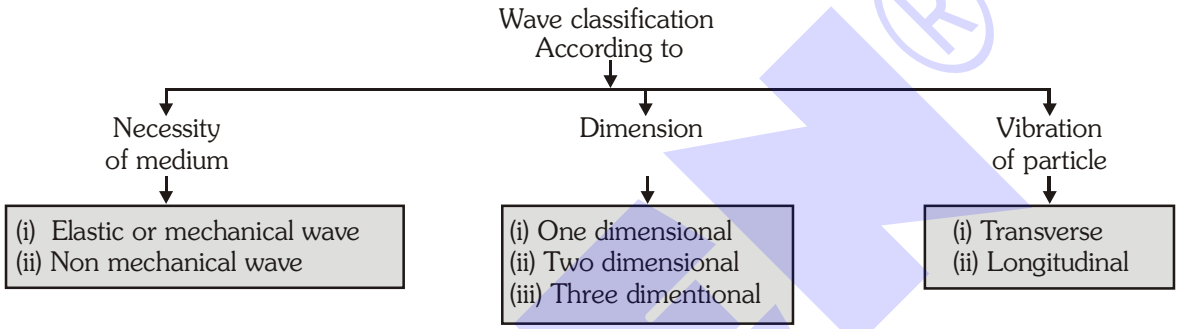
WAVE ON STRING

KEY CONCEPT

INTRODUCTION OF WAVES

A wave is a disturbance that propagates in space, transports energy and momentum from one point to another without the transport of matter.

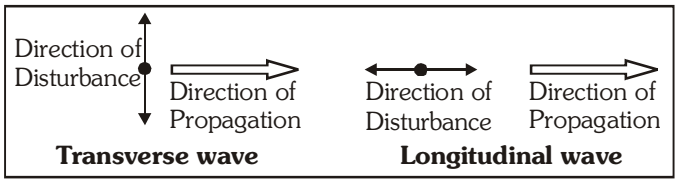
CLASSIFICATION OF WAVES



1. ***Based on medium necessity*** :- A wave may or may not require a medium for its propagation. The waves which do not require medium for their propagation are called **non-mechanical** waves, e.g. light, heat (infrared), radio waves etc.
On the other hand the waves which require medium for their propagation are called **mechanical** waves. e.g : Sound waves in water, seismic waves in earth's crust.

3. ***Based on direction of propagation*** :- Waves can be one, two or three dimensional according to the number of dimensions in which they propagate energy. Waves moving along strings are one-dimensional. Surface waves or ripples on water are two dimensional, while sound or light waves from a point source are three dimensional.

4. ***Based on the motion of particles of medium*** :
Waves are of two types on the basis of motion of particles of the medium.
(i) Longitudinal waves (ii) Transverse waves

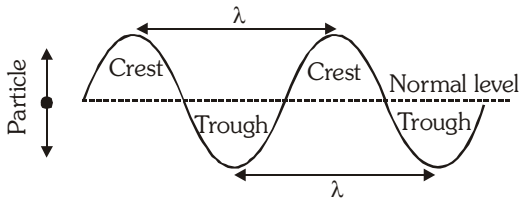


In the transverse wave the direction associated with the disturbance (i.e. motion of particles of the medium) is at right angle to the direction of propagation of wave while in the longitudinal wave the direction of disturbance is along the direction of propagation.

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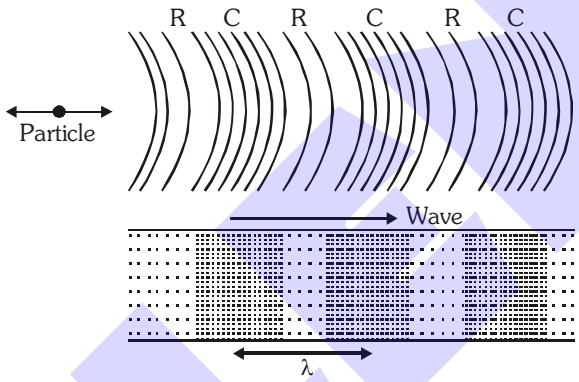
TRANSVERSE WAVE MOTION

In this individual particles of the medium execute SHM about their mean position in direction perpendicular to the direction of propagation of wave motion.



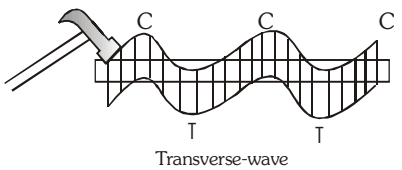
LONGITUDINAL WAVE MOTION

In this type of waves, oscillatory motion of the medium particles produces regions of compression (high pressure) and rarefaction (low pressure) which propagated in space with time (see figure).

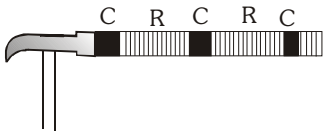


Note :

- A mechanical wave will be transverse or longitudinal depends on the nature of medium and mode of excitation.
- In strings mechanical waves are always transverse when string is under a tension. In gases and liquids mechanical waves are always longitudinal e.g. sound waves in air or water. This is because fluids cannot sustain shear.
- In solids, mechanical waves (may be sound) can be either transverse or longitudinal depending on the mode of excitation. The speed of the two waves in the same solid are different. (Longitudinal waves travels faster than transverse waves). e.g., if we struck a rod at an angle as shown in fig. (A) the waves in the rod will be transverse while if the rod is struck at the side as shown in fig. (B) or is rubbed with a cloth the waves in the rod will be longitudinal. In case of vibrating tuning fork waves in the prongs are transverse while in the stem are longitudinal.



Transverse-wave
(A)

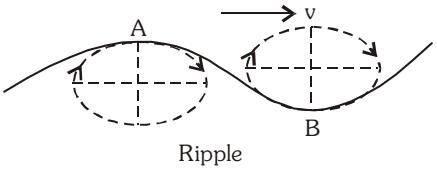


Longitudinal-wave
(B)

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Further more in case of seismic waves produced by Earthquakes both S (shear) and P (pressure) waves are produced simultaneously which travel through the rock in the crust at different speeds [$v_S \cong 5$ km/s while $v_P \cong 9$ km/s] S–waves are transverse while P–waves longitudinal.

Some waves in nature are neither transverse nor longitudinal but a combination of the two. These waves are called 'ripple' and waves on the surface of a liquid are of this type. In these waves particles of the medium vibrate up and down and back and forth simultaneously describing ellipses in a vertical plane [Fig.]



Ripple

CHARACTERISTICS OF WAVE MOTION

Some of the important characteristics of wave motion are as follows :

- In a wave motion, the disturbance travels through the medium due to repeated periodic oscillations of the particles of the medium about their mean positions.
- The energy is transferred from place to another without any actual transfer of the particles of the medium.
- Each particle receives disturbance a little later than its preceding particle i.e., there is a regular phase difference between one particle and the next.

EQUATION OF A PLANE PROGRESSIVE SINE WAVE

If, on the propagation of wave in a medium, the particles of the medium perform simple harmonic motion then the wave is called a 'simple harmonic progressive wave'. Suppose, a simple harmonic progressive wave is propagating in a medium along the positive direction of the x-axis (from left to right). In fig. (a) are shown the equilibrium positions of the particles 1, 2, 3 .....



Figure : (a)

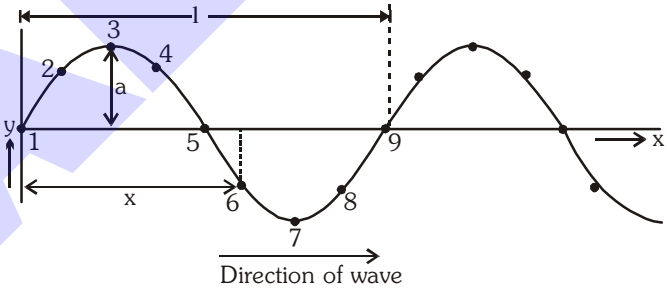


Figure : (b)

When the wave propagates, these particles oscillate about their equilibrium positions. In Fig. (b) are shown the instantaneous positions of these particles at a particular instant. The curve joining these positions represents the wave. Let the time be counted from the instant when the particle 1 situated at the origin starts oscillating. If y be the displacement of this particle after t seconds, then $y = a \sin \omega t$...(i)

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where a is the amplitude of oscillation and $\omega = 2\pi n$, where n is the frequency. As the wave reaches the particles beyond the particle 1, the particles start oscillating. If the speed of the wave be v, then it will reach particle 6, distant x from the particle 1, in x/v sec. Therefore, the particle 6 will start oscillating x/v sec after the particle 1. It means that the displacement of the particle 6 at a time t will be the same as that of the particle 1 at a time x/v sec earlier i.e. at time t – (x/v). The displacement of particle 1 at time t – (x/v) can be the particle 6, distant x from the origin (particle 1), at time t is given by

$y = a \sin \omega \left(t - \frac{x}{v}\right)$ But $\omega = 2\pi n$, $y = a \sin (\omega t - kx) \left(k = \frac{\omega}{v}\right)$ ...(ii)

$y = a \sin \left[\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right]$ Also $k = \frac{2\pi}{\lambda}$ ...(iii) $y = a \sin 2\pi \left[\frac{t}{T} - \frac{x}{\lambda}\right]$ ...(iv)

This is the equation of a simple harmonic wave travelling along +x direction. If the wave is travelling along the –x direction then inside the brackets in the above equations, instead of minus sign there will be plus sign. For example, equation (iv) will be of the following form : $y = a \sin 2\pi \left(\frac{t}{T} + \frac{x}{\lambda}\right)$. If ϕ be the phase difference between the above wave travelling along the +x direction and an other wave, then the equation of that wave will be

$$y = a \sin \left\{2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right) \pm \phi\right\}$$

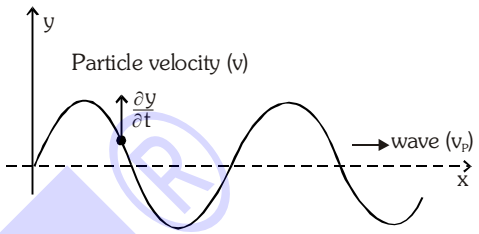
Some important terms connected with wave equation

- **Wavelength (λ) [length of one wave]**
  Distance travelled by the wave during the time, any one particle of the medium completes one vibration about its mean position. We may also define wavelength as the distance between any two nearest particles of the medium, vibrating in the same phase.
- **Frequency (n) :** Number of vibrations (Number of complete wavelengths) complete by a particle in one second.
- **Time period (T) :** Time taken by wave to travel a distance equal to one wavelength.
- **Amplitude (A) :** Maximum displacement of vibrating particle from its equilibrium position.
- **Angular frequency (ω) :** It is defined as $\omega = \frac{2\pi}{T} = 2\pi n$
- **Phase :** Phase is a quantity which contains all information related to any vibrating particle in a wave. For equation $y = A \sin (\omega t - kx)$; $(\omega t - kx)$ = phase.
- **Angular wave number (k) :** It is defined as $k = \frac{2\pi}{\lambda}$
- **Wave number ($\vec{\nu}$) :** It is defined as $\vec{\nu} = \frac{1}{\lambda} = \frac{k}{2\pi}$ = number of waves in a unit length of the wave pattern.

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- **Particle velocity, wave velocity and particle's acceleration :** In plane progressive harmonic wave particles of the medium oscillate simple harmonically about their mean position. Therefore, all the formulae what we have read in SHM apply to the particles here also. For example, maximum particle velocity is $\pm A\omega$ at mean position and it is zero at extreme positions etc. Similarly maximum particle acceleration is $\pm\omega^2 A$ at extreme positions and zero at mean position. However the wave velocity is different from the particle velocity. This depends on certain characteristics of the medium. Unlike the particle velocity which oscillates simple harmonically (between $+A\omega$ and $-A\omega$) the wave velocity is constant for given characteristics of the medium.

- **Particle velocity in wave motion :**
The individual particles which make up the medium do not travel through the medium with the waves. They simply oscillate about their equilibrium positions. The instantaneous velocity of an oscillating particle of the medium, through which a wave is travelling, is known as "Particle velocity".



- **Wave velocity :** The velocity with which the disturbance, or planes of equal phase (wave front), travel through the medium is called wave (or phase) velocity.

- **Relation between particle velocity and wave velocity :**

Wave equation :- $y = A \sin(\omega t - kx)$, Particle velocity $v = \frac{\partial y}{\partial t} = A\omega \cos(\omega t - kx)$.

Wave velocity $= v_p = \frac{\lambda}{T} = \lambda\frac{\omega}{2\pi} = \frac{\omega}{k}$,

$$\frac{\partial y}{\partial x} = -Ak\cos(\omega t - kx) = -\frac{A}{\omega}\omega k\cos(\omega k - kx) = -\frac{1}{v_p}\frac{\partial y}{\partial t}$$

$$\Rightarrow \frac{\partial y}{\partial x} = -\frac{1}{v_P}\frac{\partial y}{\partial t}$$

Note : $\frac{\partial y}{\partial x}$ represent the slope of the string (wave) at the point x.

Particle velocity at a given position and time is equal to negative of the product of wave velocity with slope of the wave at that point at that instant.

- **Differential equation of harmonic progressive waves :**

$$\frac{\partial^2 y}{\partial t^2} = -A\omega^2\sin(\omega t - kx) \Rightarrow \frac{\partial^2 y}{\partial x^2} = -Ak^2\sin(\omega t - kx) \Rightarrow \boxed{\frac{\partial^2 y}{\partial x^2} = \frac{1}{v_p^2}\frac{\partial^2 y}{\partial t^2}}$$

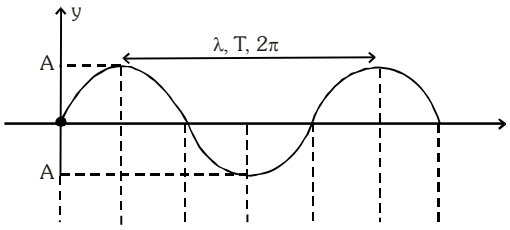
- **Particle velocity (v_p) and acceleration (a_p) in a sinusoidal wave :**
The acceleration of the particle is the second particle is the second partial derivative of y (x, t) with respect to t,

$$\therefore\ a_P = \frac{\partial^2 y(x,t)}{\partial t^2} = \omega^2 A\sin(kx - \omega t) = -\omega^2 y(x,t)$$

i.e., the acceleration of the particle equals $-\omega^2$ times its displacement, which is the result we obtained for SHM. Thus, $a_p = -\omega^2$ (displacement)

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- **Relation between Phase difference, Path difference & Time difference**

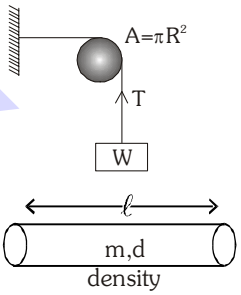


Phase (ϕ)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$	3π
Wave length (λ)	0	$\frac{\lambda}{4}$	$\frac{\lambda}{2}$	$\frac{3\lambda}{4}$	λ	$\frac{5\lambda}{4}$	$\frac{3}{2}\lambda$
Time-period (T)	0	$\frac{T}{4}$	$\frac{T}{2}$	$\frac{3T}{4}$	T	$\frac{5T}{4}$	$\frac{3T}{2}$

$\Rightarrow \frac{\Delta\phi}{2\pi} = \frac{\Delta\lambda}{\lambda} = \frac{\Delta T}{T} \Rightarrow$ Path difference $= \left(\frac{\lambda}{2\pi}\right)$ Phase difference

VELOCITY OF TRANSVERSE WAVE

Mass of per unit length $\mu = \frac{\pi r^2 \ell \times d}{\ell}$, $\mu = \pi r^2 d$, where d = Density of matter



Velocity of transverse wave in any wire $v = \sqrt{\frac{T}{\mu}}$ or $\sqrt{\frac{T}{\pi r^2 d}} = \sqrt{\frac{T}{Ad}} \because \pi r^2 = A$

- If m is constant then, $v \propto \sqrt{T}$ it is called tension law.
- If tension is constant then $v \propto \sqrt{\frac{1}{\mu}}$ ← it is called law of mass
- If T is constant & take wire of different radius for same material then $v \propto \frac{1}{r}$ ← it is called law of radius
- If T is constant & take wire of same radius for different material. Then $v \propto \sqrt{\frac{1}{d}}$ ← law of density

ENERGY DENSITY AND POWER IN A TRAVELLING WAVE

Energy density in a travelling wave

Kinetic energy density $= \frac{\partial K}{\partial x} = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2$

Potential energy density $= \frac{\partial U}{\partial x} = \frac{1}{2}T\left(\frac{\partial y}{\partial x}\right)^2$

Total energy density $= \frac{\partial E}{\partial x} = \frac{\partial K}{\partial x} + \frac{\partial U}{\partial x}$

For a travelling wave, $\frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}$

$\therefore \frac{\partial K}{\partial x} = \frac{\partial U}{\partial x}$

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or $\frac{\partial E}{\partial x} = \frac{2\partial K}{\partial x} = \frac{2\partial U}{\partial x} = \mu\left(\frac{\partial y}{\partial t}\right)^2$

for a sine wave, $y = A \sin(\omega t - kx + \phi)$

$$\frac{\partial E}{\partial x} = \mu\left(A\omega\cos\left(\omega t - kx + \phi\right)\right)^2$$

$$\frac{\partial E}{\partial x} = \mu A^2\omega^2 \cos^2\left(\omega t - kx + \phi\right)$$

$$\left.\frac{\partial E}{\partial x}\right|_{avg} = \frac{\mu A^2\omega^2}{2}$$

Energy in one wavelength $= \left.\frac{\partial E}{\partial x}\right|_{avg} \times \lambda$

$$= \frac{\mu A^2\omega^2}{2}\cdot\lambda$$

Average power transferred by the wave

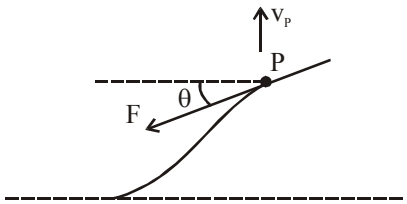
$$= \frac{E_{avg}}{T} = \frac{\mu A^2\omega^2}{2}\cdot\frac{\lambda}{T} = \frac{\mu v A^2\omega^2}{2}$$

Power transmitted along the string

$F_y = -F\sin\theta$

$P = (-F\sin\theta)\cdot v_P$

$\simeq -F\tan\theta\cdot v_P$



$$P = -F\frac{\partial y}{\partial x}\cdot\frac{\partial y}{\partial t}$$

for a sine wave, $y = A \sin(\omega t - kx + \phi)$

$P = +F(A\omega\cos(\omega t - kx + \phi)\cdot Ak\cos(\omega t - kx + \phi)$

$P = +F(A^2\omega k)\cos^2(\omega t - kx + \phi)$

$$= +\mu v^2 A^2\frac{\omega^2}{v}\cos^2\left(\omega t - kx + \phi\right)$$

$P = \mu v A^2\omega^2\cos^2(\omega t - kx + \phi)$

$$P_{avg} = \frac{\mu v A^2\omega^2}{2}$$

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SUPERPOSITION PRINCIPLE

Two or more waves can propagate in the same medium without affecting the motion of one another. If several waves propagate in a medium simultaneously, then the resultant displacement of any particle of the medium at any instant is equal to the vector sum of the displacements produced by individual wave. The phenomenon of intermixing of two or more waves to produce a new wave is called Superposition of waves. Therefore according to superposition principle.

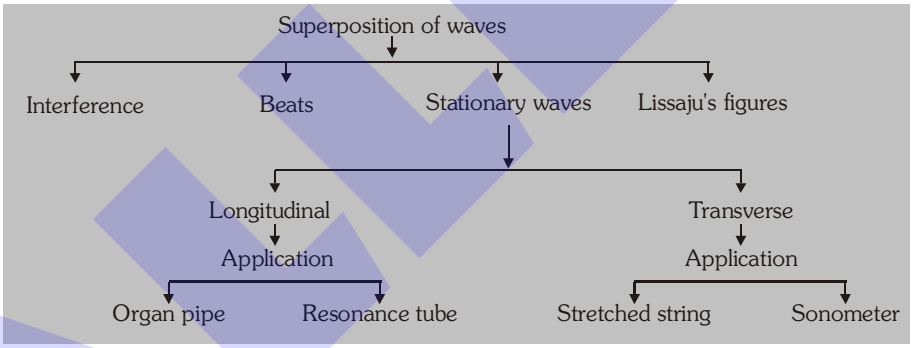
> The resultant displacement of a particle at any point of the medium, at any instant of time is the vector sum of the displacements caused to the particle by the individual waves.

If $\vec{y}_1, \vec{y}_2, \vec{y}_3, \ldots$ are the displacement of particle at a particular time due to individual waves, then the resultant displacement is given by $\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \ldots$

Principle of superposition holds for all types of waves, i.e., mechanical as well as electromagnetic waves. But this principle is not applicable to the waves of very large amplitude.

Due to superposition of waves the following phenomenon can be seen

- **Interference :** Superposition of two waves having equal frequency and nearly equal amplitude.
- **Beats :** Superposition of two waves of nearly equal frequency in same direction.
- **Stationary waves :** Superposition of equal wave from opposite direction.
- **Lissajous' figure :** Superposition of perpendicular waves.



REFLECTION AND TRANSMISSION OF WAVES

1. Reflection from rigid end

When the pulse reaches the right end which is clamped at the wall, the element at the right end exerts a force on the clamp and the clamp exerts equal and opposite force on the element. The element at the right end is thus acted upon by the force from the clamp. As this end remains fixed, the two forces are opposite to each other. The force from the left part of the string transmits the forward wave pulse and hence, the force exerted by the clamp sends a return pulse on the string whose shape is similar to a return pulse but is inverted. The original pulse tries to pull the element at the fixed end up and the return pulse sent by the clamp tries to pull it down, so the resultant displacement is zero. Thus, the wave is reflected from the fixed end and the reflected wave is inverted with respect to the original wave. The shape of the string at any time, while the pulse is being reflected, can be found by adding an inverted image pulse to the incident pulse.

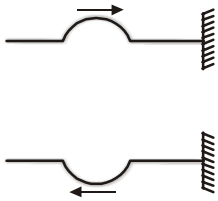
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Equation of wave propagating in +ve x-axis

Incident wave $y_1 = a \sin(\omega t - kx)$

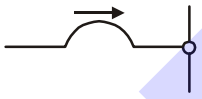
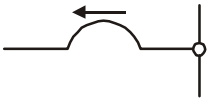
Reflected wave $y_2 = a \sin(\omega t + kx + \pi)$

or $y_2 = -a \sin(\omega t + kx)$



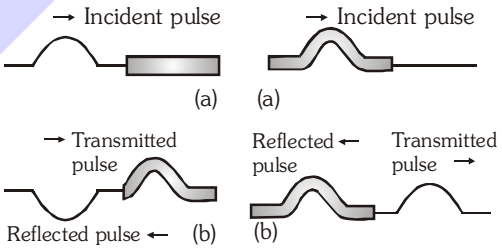
2. Reflection from free end

The right end of the string is attached to a light frictionless ring which can freely move on a vertical rod. A wave pulse is sent on the string from left. When the wave reaches the right end, the element at this end is acted on by the force from the left to go up. However, there is no corresponding restoring force from the right as the rod does not exert a vertical force on the ring. As a result, the right end is displaced in upward direction more than the height of the pulse i.e., it overshoots the normal maximum displacement. The lack of restoring force from right can be equivalent described in the following way. An extra force acts from right which sends a wave from right to left with its shape identical to the original one. The element at the end is acted upon by both the incident and the reflected wave and the displacements add. Thus, a wave is reflected by the free end without inversion.

Incident wave $y_1 = a \sin(\omega t - kx)$  Reflected wave $y_2 = a \sin(\omega t + kx)$ 

3. Transmission of waves

We may have a situation in which the boundary is intermediate between these two extreme cases, that is, one in which the boundary is neither rigid nor free. In this case, part of the incident energy is transmitted and part is reflected. For instance, suppose a light string is attached to a heavier string as in (figure). When a pulse travelling on the light reaches the knot, same part of it is reflected and inverted and same part of it is transmitted to the heavier string.



As one would expect, the reflected pulse has a smaller amplitude than the incident pulse, since part of the incident energy is transferred to the pulse in the heavier string. The inversion in the reflected wave is similar to the behaviour of a pulse meeting a rigid boundary, when it is totally reflected. When a pulse travelling on a heavy string strikes the boundary of a lighter string, as in (figure), again part is reflected and part is transmitted. However, in this case the reflected pulse is not inverted. The following general rules apply to reflected waves. When a wave pulse travels from medium A to medium B and $v_A > v_B$ (that is, when B is denser than A), the pulse will be inverted upon reflection. When a wave pulse travels from medium A to medium B and $v_A < v_B$ (A is denser than B), it will not be inverted upon reflection. The amplitude of reflected wave and transmitted wave is given by

$$A_r = \left(\frac{v_B - v_A}{v_B + v_A}\right) A_i, \quad A_t = \frac{2v_B}{v_A + v_B} A_i$$

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STATIONARY WAVES

* **Definition :** The wave propagating in such a medium will be reflected at the boundary and produce a wave of the same kind travelling in the opposite direction. The superposition of the two waves will give rise to a stationary wave. Formation of stationary wave is possible only and only in bounded medium.

- **From rigid end** : We know equation for progressive wave in positive x-direction $y_1 = a \sin(\omega t - kx)$
  After reflection from rigid end $y_2 = a \sin(\omega t + kx + \pi) = -a \sin(\omega t + kx)$
  By principle of super position. $y = y_1 + y_2 = a \sin(\omega t - kx) - a \sin(\omega t + kx) = -2a \sin kx \cos \omega t$
  This is equation of stationary wave reflected from rigid end

  Amplitude = 2a sin kx **Velocity of particle** $v_{pa} = \frac{dy}{dt} = 2a\,\omega \sin kx \sin \omega t$

  Node $x = 0, \frac{\lambda}{2}, \lambda$ .....

  Antinode $x = \frac{\lambda}{4}, \frac{3\lambda}{4}$ .....

- **From free end** : we know equation for progressive wave in positive x-direction $y_1 = a \sin(\omega t - kx)$
  After reflection from free end $y_2 = a \sin(\omega t + kx)$
  By Principle of superposition $y = y_1 + y_2 = a \sin(\omega t - kx) + a \sin(\omega t + kx) = 2\ a \sin \omega t \cos kx$
  Amplitude = 2a cos kx,

  - **Antinode :** $x = 0, \frac{\lambda}{2}, \lambda$ ..... $A \rightarrow$ Max,
  - **Node :** $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ ..... A = 0,

PROPERTIES OF STATIONARY WAVES

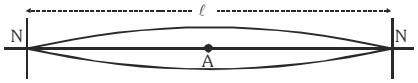
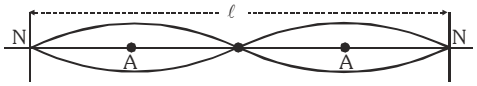
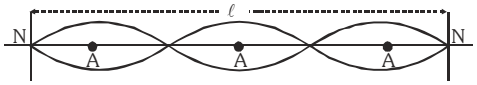
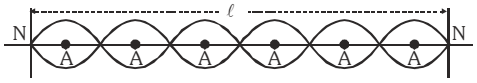
The stationary waves are formed due to the superposition of two identical simple harmonic waves travelling in opposite direction with the same speed.

Important characteristics of stationary waves are:–

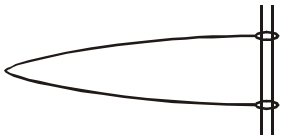
(i) Stationary waves are produced in the bounded medium and the boundaries of bounded medium may be rigid or free.
(ii) In stationary waves nodes and antinodes are formed alternately. Nodes are the points which are always in rest having maximum strain. Antinodes are the points where the particles vibrate with maximum amplitude having minimum strain.
(iii) All the particles except at the nodes vibrate simple harmonically with the same period.
(iv) The distance between any two successive nodes or antinodes is $\lambda/2$.
(v) The amplitude of vibration gradually increases from zero to maximum value from node to antinode.
(vi) All the particles in one particular segment vibrate in the same phase, but the particle of two adjacent segments differ in phase by 180°

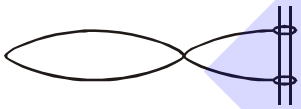
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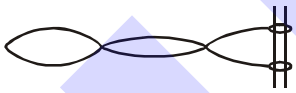
(i) Transverse Stationary wave (fixed at both ends)

First Harmonic/Fundamental		$\frac{\lambda}{2}=\ell \Rightarrow \lambda=2\ell \Rightarrow f=\frac{v}{2\ell}$
Second Harmonic		$\frac{2\lambda}{2}=\ell \Rightarrow \lambda=\ell \Rightarrow f=\frac{v}{\ell}$
Third Harmonic		$\frac{3\lambda}{2}=\ell \Rightarrow \lambda=\frac{2\ell}{3} \Rightarrow f=\frac{3v}{2\ell}$
p^{th} harmonic		$\frac{p\lambda}{2}=\ell \Rightarrow \lambda=\frac{2\ell}{p} \Rightarrow f=\frac{pv}{2\ell}$

(ii) Transverse stationary wave (fixed at one end and free at one end)

Fundamental  $\frac{\lambda}{4}=\ell \Rightarrow \lambda=4\ell, f=\frac{v}{4\ell}$

Third harmonic  $\frac{3\lambda}{4}=\ell \Rightarrow \lambda=\frac{4\ell}{3}, f=\frac{3v}{4\ell}$

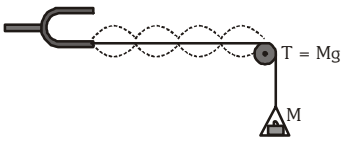
Fifth harmonic  $\frac{5\lambda}{4}=\ell \Rightarrow \lambda=\frac{4\ell}{5}, f=\frac{5v}{4\ell}$

$(2n-1)^{th}$ harmonic $f=\frac{(2n-1)v}{4\ell}$

MELDE'S EXPERIMENT :

In Melde's experiment, one end of a flexible piece of thread is tied to the end of a tuning fork. The other end passed over a smooth pulley carries a pan which can be loaded. There are two arrangements to vibrate the tied fork with thread.

Transverse arrangement :



Case 1. In a vibrating string of fixed length, the product of number of loops and square root of tension are constant or $p\sqrt{T}$ = constant.

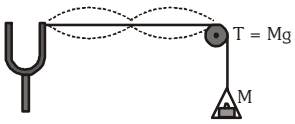
Case 2. When the tuning fork is set vibrating as shown in fig. then the prong vibrates at right angles to the thread. As a result the thread is set into motion. The frequency of vibration of the thread (string) is equal to the frequency of the tuning fork. If length and tension are properly adjusted then, standing waves are formed in the string. (This happens when frequency of one of the normal modes of the string matched with the frequency of the tuning fork). Then, if p loops are formed in the thread, then the frequency of the tuning fork is given by $n=\frac{p}{2\ell}\sqrt{\frac{T}{m}}$

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Case 3. If the tuning fork is turned through a right angle, so that the prong vibrates along the length of the thread, then the string performs only a half oscillation for each complete vibrations of the prong. This is because the thread only makes node at the midpoint when the prong moves towards the pulley i.e. only once in a vibration.

Longitudinal arrangement :

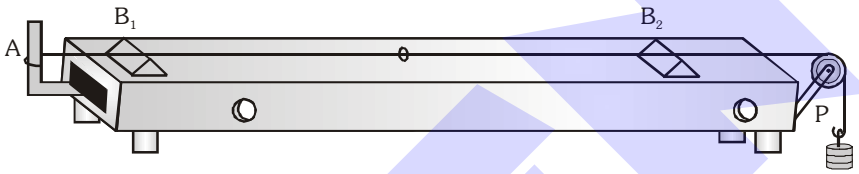
The thread performs sustained oscillations when the natural frequency of the given length of the thread under tension is half that of the fork.



Thus if p loops are formed in the thread, then the frequency of the tuning fork is $n = \frac{2p}{2\ell}\sqrt{\frac{T}{m}}$

SONOMETER :

Sonometer consists of a hollow rectangular box of light wood. One end of the experimental wire is fastened to one end of the box. The wire passes over a frictionless pulley P at the other end of the box. The wire is stretched by a tension T.



The box serves the purpose of increasing the loudness of the sound produced by the vibrating wire. If the length the wire between the two bridges is ℓ, then the frequency of vibration is $n = \frac{1}{2\ell}\sqrt{\frac{T}{m}}$

To test the tension of a tuning fork and string, a small paper rider is placed on the string. When a vibrating tuning fork is placed on the box, and if the length between the bridges is properly adjusted, then when the two frequencies are exactly equal, the string quickly picks up the vibrations of the fork and the rider is thrown off the wire. There are three laws of vibration of a wire.

COMPARISON OF PROGRESSIVE AND STATIONARY WAVES

	Progressive waves	**Stationary waves**
1.	These waves travels in a medium with definite velocity.	These waves do not travel and remain confined between two boundaries in the medium.
2.	These waves transmit energy in the medium.	These waves do not transmit energy in the medium.
3.	The phase of vibration varies continuously from particle to particle.	The phase of all the particles in between two nodes is always same. But particles of two Adjacent nodes differ in phase by 180°
4.	No particle of medium is Permanently at rest.	Particles at nodes are permanently at rest.
5.	All particles of the medium vibrate and amplitude of vibration is same.	The amplitude of vibration changes from particle to particle. The amplitude is zero for all at nodes and maximum at antinodes.
6.	All the particles do not attain the maximum displacement position simultaneously.	All the particles attain the maximum displacement

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